1	Introduction to Statistics	
1	Descriptive vs inferential statistics	
2	Population vs sample	
3	Parameters vs statistics	
4	Qualitative vs quantitative data	
5	Levels of measurement (nominal, ordinal, interval, ratio)	
6	Discrete vs continuous variables	
7	Cross-sectional vs time-series data	
8	Structured vs unstructured data	
9	Data collection methods	
10	Common data sources in data science	

2	Measures of Central Tendency	
1	Arithmetic mean	
2	Weighted mean	
3	Geometric and harmonic mean	
4	Median	
5	Mode	
6	Trimmed mean	
7	Percentiles	
8	Quartiles	
9	Five-number summary	
10	Effect of outliers on central tendency	

3	Measures of Spread	
1	Range	
2	Interquartile range (IQR)	
3	Variance (population and sample)	
4	Standard deviation	
5	Mean absolute deviation	
6	Coefficient of variation	
7	Skewness	
8	Kurtosis	
9	Standardization vs normalization	
10	Empirical rule (68-95-99.7)	

4	Data Visualization	
1	Frequency distributions	
2	Histograms	
3	Bar charts and pie charts	
4	Box plots and violin plots	
5	Scatter plots	
6	Line charts	
7	Q-Q plots	
8	Density plots and kernel density estimation	
9	Heatmaps	
10	Pair plots	

5	Sampling Distributions	
1	Simple random sampling	
2	Stratified sampling	
3	Cluster sampling	
4	Systematic sampling	
5	Convenience and judgment sampling	
6	Sampling bias	
7	Sample size determination	
8	Sampling distribution of the mean	
9	Standard error	
10	Imbalanced data and resampling techniques	

6	Confidence Intervals	
1	Point estimation	
2	Interval Estimation	
3	Maximum likelihood estimation	
4	Bias – Variance Tradeoff	
5	Confidence intervals for the mean (z and t)	
6	Confidence intervals for proportions	
7	Confidence intervals for variance	
8	Margin of error	
9	Interpreting confidence intervals	
10	Common misinterpretations of confidence intervals	

7	AB Testing - Hypothesis Testing	
1	Null & Alternative hypotheses	
2	Control Group &Treatment Group	
3	Test statistics	
4	Significance level (alpha)	
5	p-values	
6	Type I and Type II errors	
7	Statistical power and effect size	
8	Z-test (one-sample and two-sample)	
9	t-test (one-sample, two-sample, paired)	
10	One-tailed vs two-tailed tests	

8	Inferential Statistics	
1	One-way ANOVA	
2	Two-way ANOVA	
3	F-test for comparing variances	
4	Correlation is not Causation	
5	Latent Variables	
6	Martingale Property	
7	Markow Chain Property	
8	Monte Carlo Simulation	
9	Randomized Control Trials - RCTs	
10	Interaction Effects	

9	Correlation and Covariance	
1	Covariance	
2	Pearson correlation coefficient	
3	Spearman rank correlation	
4	Correlation vs causation	
5	Spurious correlation	
6	Confounding variables	
7	Simpson's paradox	
8	Interaction effects	
9	Law of Large Numbers	
10	Central Limit Theorem	

10	Outlier Detection	
1	Z-score standardization	
2	Min-max normalization	
3	Log and power transformations	
4	Handling missing data	
5	Encoding categorical variables (one-hot, label encoding)	
6	Z-score → classic baseline for outliers	
7	IQR / Tukey Rule → simple, no strong assumptions	
8	Modified Z-score skewed or dirty data	
9	Mahalanobis distance → multiple numeric columns	
10	Minimum Covariance Determinant MCD → dirty data with outliers	